Israel-Stewart Causal Viscous Perturbations in the SSEE Dark Energy Framework: Stability from the Field, Weak Dark-Energy Clustering, and ΛCDM-Consistent Structure Growth

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

November 2026 (SSEE, Paper 5 of 10)

Abstract

The Structural Self-Energy Expansion (SSEE) derives a dark energy equation of state $w_0 = -T_r/M_v = -0.839950$ and $w_a = -0.669975$ purely from the golden ratio φ and π, with zero fitted dimensionless parameters in the dark-energy background sector, in agreement with the official DESI DR2 w_0w_aCDM posterior at 0.24σ (Pantheon+; 1.5–1.8σ across SN compilations). (Following Paper 1 §1.3, the broader SSEE programme carries a small set of phenomenological ansätze in the dark-matter sector — in particular the ϕ-DM mass and the non-minimal coupling ξ — retracted with the φ-DM sector (Paper 6); OP-9/11 closed by dissolution, OP-14 resolved in `OPEN_PROBLEMS.md`; the present perturbation analysis is independent of those ansätze.) Because $w_0 < -1/3$, a naive k-essence interpretation gives a negative bare sound speed $c^2_{s,\mathrm{bare}} = w_0 = -0.839950$, implying catastrophic gradient instability on all sub-horizon scales.

We perform a linear Israel-Stewart (IS) causal viscous perturbation analysis of SSEE dark energy in the sub-horizon regime, addressing three interconnected questions: *(Q1)* Does IS regularise the gradient instability for all observationally relevant modes? *(Q2)* Do linear IS perturbations dynamically source the factor ≈ 2 that earlier drafts read between $\Omega_{m,\mathrm{CMB}}$ and the number $s_m = 0.160050$, or is there nothing to source (the former interpretation as a "MIRA factor" having been retired in the ω_m-direct reframe, OP-8 dissolved, and s_m being a quantity of the equation of state rather than a second matter sector)? *(Q3)* What are the falsifiable predictions for the growth rate, σ_8, and the S_8 parameter?

For Q1, the SSEE structure constants satisfy the algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = s_{\mathrm{DE}} = |w_0|$ (the saturation $s_{\mathrm{DE}} = T_r/M_v$ of Paper 1), so that the high-frequency ($\omega\tau_\Pi \to \infty$) sound speed of the IS fluid vanishes, $c^2_{s,\mathrm{eff}} = w_0 + s_{\mathrm{DE}} = 0$. We show that this is *not* a stability result. For a mode growing at rate λ the viscous stiffness is $s_{\mathrm{DE}}\,\lambda\tau_\Pi/(1+\lambda\tau_\Pi) < s_{\mathrm{DE}}$, so with $c^2_{s,\mathrm{bare}} = w_0$ the growing branch obeys $\lambda^2 = k^2 s_{\mathrm{DE}}/(1+\lambda\tau_\Pi) > 0$ at every k ($\lambda = 15.5\,H_0$ at $k = 100\,H_0/c$); the full

five-variable integration confirms it. The stability of the SSEE dark-energy sector is carried instead by the field of Paper 7, whose own sound speed is $c_s^2 = +0.021284$: with it, and the same viscosity, no mode grows. The quasi-static reduction used for this question in earlier versions is the Navier–Stokes limit, valid only for $\tau_\Pi H \ll 1$, which never holds here.

For Q2, integrating the full linear IS system (with the field's sound speed) from $z = 19$ to $z = 0$ gives $\delta_{\rm DE}/\delta_m = 1.5 \times 10^{-3}$ at $k = 10\,H_0/c$, falling as k^{-2}: the enhancement $\Omega_{m,\rm eff}/\Omega_m = 1.0010 \pm 0.0013$ ($k \geq 10\,H_0/c$) is a bound on dark-energy clustering, not a factor ≈ 2 — and there is no second matter sector for it to supply. The number $s_m = 0.160050$ once paired with $\Omega_{m,\rm CMB}$ is not a density but $1+w_0$, a quantity of the equation of state, so the difference $0.308881 - 0.160050$ that once defined a second sector subtracts objects of different kinds; that construction is retracted (Paper 6). The matter content is a single sector, $\Omega_m = \omega_m/h^2 = 0.308881$, and the algebraic number MIRA $= (3\varphi + \pi)/4 = 1.998924$ survives only as the Hubble-screening amplitude of Paper 9. The same integration, written in the regular variable of Kunz & Sapone, carries the perturbations through the phantom crossing at $z_* = 0.314$ without divergence.

For Q3, we compute the IS growth index $\gamma_{\rm IS} = 0.5504 \pm 0.0003$ (consistent with $\gamma_{\Lambda\rm CDM} \simeq 0.55$), the growth factor $\mathcal{G} = D_1^{\rm SSEE}/D_1^{\Lambda\rm CDM} = 1.0032$ (growth essentially identical to ΛCDM once the correct cosmological background $\Omega_{m,\rm cosm} = 0.308881$ is used in the Poisson source; see Paper 1, §"ω_m-direct" — the Two-Ω_m Criterion that once stated this rule is itself *withdrawn*, because $s_m = 0.160050$ was never a density that could compete for the slot), and the predicted $S_8 = \sigma_8(\Omega_{m,\rm CMB}/0.3)^{0.5}$. The amplitude is a separate matter and is not this paper's result. Fixing A_s to Planck gives $S_8^{\rm SSEE} = 0.8256 \pm 0.0061$, and the conservative CLASS ceiling is higher still, $S_8^{\rm ceiling} = 0.827$. Both are *ceilings under an imported normalisation*, not predictions: A_s is one of the two free CMB parameters of the model, and fixing it to Planck imports the Planck–KiDS offset instead of predicting it. With A_s free and this same background, against the raw KiDS-Legacy $\xi_\pm$ (357 points) the amplitude the shear asks for lies -0.25σ from the one the model's own CMB fit supplies, so A_s can be fixed there as well, and with *no* free cosmological parameter the model predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured (Paper 6).[1] What this paper establishes is the growth *shape*, $\mathcal{G}$ and $\gamma_{\rm IS}$, which are amplitude-independent. The $f\sigma_8$ mean tension is 0.70σ, essentially identical to ΛCDM (0.73σ): SSEE does not suppress or enhance structure formation relative to ΛCDM once $\Omega_{m,\rm cosm} = 0.308881$ governs the growth equation.

Finally, we place the IS layer in the EFT-of-dark-energy language [Gubitosi et al., 2013, Bellini and Sawicki, 2014]: the dark-energy action of Paper 7 is a shift-symmetric ghost condensate with $\alpha_T = \alpha_B = \alpha_M = 0$, which produces no bulk viscosity, so the IS layer is a property of the fluid parametrisation and maps to no operator of the action. An earlier mapping to $\bar{M}_2^2$ (which would imply $\alpha_T \neq 0$) and an earlier contrast with the ghost condensate ("$c_s^2 = 1$") are withdrawn: the ghost condensate *is* the SSEE dark-energy sector.

[1] KiDS-Legacy (Wright et al. 2025, arXiv:2503.19441) was published sixteen months before this analysis; no temporal priority is claimed. Its recalibration of the source redshift distributions dissolves the Planck–shear disagreement in the data themselves: re-running the KiDS-1000 analysis under blinding, the collaboration's Appendix I finds $2.39\sigma \to 1.42\sigma$ from the $n(z)$ calibration alone. The three χ^2 compared span 0.74 over 357 points, so the shear does *not* discriminate between the models; what separates them is the parameter count. See Paper 6.

Contents

1 Introduction **4**
- 1.1 The SSEE framework and the perturbation challenge . . . 4
- 1.2 The gradient instability problem . . . 4
- 1.3 The Israel–Stewart layer . . . 5
- 1.4 Paper structure . . . 6

2 Israel-Stewart Theory for SSEE Dark Energy **6**
- 2.1 Entropy production and the second-order IS equations . . . 6
- 2.2 SSEE IS parameters from algebraic structure constants . . . 7
- 2.3 Thermodynamic consistency . . . 9
- 2.4 Linearised perturbation equations in Newtonian gauge . . . 9
- 2.5 The quasi-static reduction is the Navier–Stokes limit . . . 10

3 Q1: Gradient Stability — What the IS Identity Does and Does Not Do **10**
- 3.1 IS dispersion relation from linear perturbation theory . . . 10
- 3.2 The high-frequency identity . . . 11
- 3.3 The growing branch: the closure $c_s^2 = w_0$ stays unstable . . . 11
- 3.4 Uniqueness of the minimal-viscosity solution . . . 11
- 3.5 Critical wavenumber and observational window . . . 12

4 Q2: How Much Does Dark Energy Cluster? **12**
- 4.1 Numerical setup and initial conditions . . . 12
- 4.2 The scale that a two-fold enhancement would need . . . 13
- 4.3 Numerical results . . . 13
- 4.4 Why the ratio falls as k^{-2} . . . 14
- 4.5 Through the phantom crossing . . . 14

5 Q3: Growth Rate, σ_8, and the S_8 Tension **15**
- 5.1 Linear growth factor and growth rate . . . 15
- 5.2 Numerical results for the SSEE growth index . . . 15
- 5.3 Linear growth factor and σ_8 . . . 16
- 5.4 S_8 prediction and the weak-lensing tension . . . 16
- 5.5 Comparison with RSD surveys: $f\sigma_8(z)$ . . . 18

6 EFT Correspondence **19**

7 Discussion **20**
- 7.1 The withdrawn CMB-to-s_m ratio (historical) . . . 20
- 7.2 Implications for JWST early galaxy counts . . . 21
- 7.3 The S_8–$f\sigma_8$ alignment and the role of $\Omega_{m,\mathrm{cosm}}$ . . . 22
- 7.4 The H_0 tension and SSEE . . . 22
- 7.5 Causal structure and Hiscock-Lindblom stability . . . 22
- 7.6 Falsifiable predictions summary . . . 23

8 Conclusions **23**

A Derivation of the IS Dispersion Relation **25**

B Eigenvalue Analysis of the IS Perturbation Matrix **26**

1 Introduction

1.1 The SSEE framework and the perturbation challenge

The Structural Self-Energy Expansion (SSEE) [Almeida Vallejo, 2026a] is a minimal-parameter dark energy framework (zero fitted dimensionless parameters in the background sector; see Paper 1 §1.3) whose equation-of-state parameters

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi+\beta)}{\varphi+\pi+K_v} = -0.839950, \qquad w_a = -\frac{P_{sc}}{I_g} = -0.669975, \tag{1}$$

are derived algebraically from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π, without any continuous free parameters. Here $\beta = (\pi+\varphi)/2$, $K_v = \varphi+\pi+(\pi+\varphi)$, $I_g = K_v$ (the structural constraint of Paper 1), $T_r = 3(\varphi+\beta)$, $M_v = \varphi+\pi+K_v$, and $P_{sc} = (\pi+\varphi)+\varphi$. The CPL parameters (1) [Chevallier and Polarski, 2001, Linder, 2003] are consistent with DESI DR2 at 0.24σ (Pantheon+) [Almeida Vallejo, 2026c, Abdul-Karim et al., 2025] and with the Planck 2018 CMB spectra [Almeida Vallejo, 2026d].

Paper 4 [Almeida Vallejo, 2026e] identifies the algebraic structure constants

$$\mathrm{KAL}_0 \equiv \beta+\pi \simeq 5.5214 \quad \text{(Structural Retention)}, \tag{2}$$

$$\mathrm{MIRA} \equiv \frac{3\varphi+\pi}{4} \simeq 1.9989 \quad \text{(screening amplitude, Paper 9)}, \tag{3}$$

and (under the ω_m-direct reframe; OP-8 dissolved) the CMB matter density is the standard physical $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 \simeq 0.308881$, built from the algebraic $\omega_m = \omega_b+\omega_c+\omega_\nu$, in 0.88σ agreement with Planck 2018. (The earlier MIRA mapping $s_m \times \mathrm{MIRA}$ is superseded.)

The present paper confronts a fundamental question that must be answered before SSEE can claim theoretical consistency: *does the dark energy sector remain perturbatively stable*?

1.2 The gradient instability problem

A single-power (constant-w) k-essence proxy with Lagrangian $\mathcal{L} \propto X^n$ [Armendariz-Picon et al., 2001], where $X = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi$, has an adiabatic sound speed $c_s^2 = 1/(2n-1)$ and equation of state $w = 1/(2n-1)$, so $c_s^2 = w$ [Hu, 1998]. For SSEE this gives

$$c^2_{s,\mathrm{bare}} = w_0 = -0.839950. \tag{4}$$

Negative c_s^2 signals a *gradient instability*: perturbations $\delta_{\mathrm{DE}} \propto e^{|c_s|kt}$ grow exponentially on a timescale $(|c_s|k)^{-1}$, which for $k \gtrsim 10\,H_0/c$ (BAO scales) is shorter than a Hubble time [Caldwell, 2002]. Left unchecked, this instability would generate excessive DE density contrast $|\delta_{\mathrm{DE}}| \gg 1$, invalidating linear perturbation theory and producing observable signatures in the CMB lensing and matter power spectra inconsistent with data [Calabrese et al., 2011].

We stress that this bare instability is an artefact of the *effective-fluid* (constant-w) closure $c_s^2 = w$, *not* a property of the fundamental SSEE field. The dark-energy action is the ghost condensate of Paper 7, $K = c_1X + c_2X^2$ with $c_2X/c_1 = -0.522735$, which is independently ghost-free and gradient-stable with adiabatic sound speed

$$c^2_{s,\mathrm{ad}} = \frac{1+2u}{1+6u} = \frac{M_v - T_r}{5M_v + 3T_r} = 0.021284, \qquad u \equiv c_2X/c_1, \tag{5}$$

a clustered rather than smooth dark energy. *Correction (2026-09-06).* Earlier versions of this paragraph quoted $c^2_{s,\mathrm{ad}} \in [0.60, 1]$ and attributed the Lagrangian to "Papers 7 and 10". That value was computed from $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$, which is Paper 10's *screening* functional and not the dark-energy action: the two carry different coefficient ratios ($+0.008577$ against -0.522735) and do different jobs [Almeida Vallejo, 2026b]. The sound speed of the dark-energy sector is the one above. The IS analysis below addresses the *hydrodynamic* (fluid) description used by Boltzmann codes, where dark energy is parametrised by (w, c_s^2) and the naive closure $c_s^2 = w$ would be catastrophic. The IS bulk viscosity is $\tilde{\zeta} = \mathrm{KAL}_0/3$, so the fluid-level layer introduces no new parameter. We show in Section 3 that this layer does *not*, by itself, remove the instability of the closure $c_s^2 = w$; what removes it is the field's own sound speed above.

What is and is not fixed here (2026-09-07). Only the *ratio* enters the observable: $\tilde{\zeta}/(\tau_\Pi H_0) = s_{\mathrm{DE}}$, and since $\tilde{\zeta}$ and τ_Π share the factor KAL_0, it cancels. The high-frequency result $c^2_{s,\mathrm{eff}} = w_0 + s_{\mathrm{DE}} = 0$ is therefore *independent of* KAL_0: replacing it by any positive constant returns the same 0, which follows from the algebraic identity $s_{\mathrm{DE}} = |w_0|$ alone. The individual split $\tilde{\zeta} = \mathrm{KAL}_0/3$ is consequently not determined by anything in *this* observable, and is not claimed to be.

It does not follow that the IS constants are unconstrained. KAL_0 enters the framework in three places that do move observables: $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ (a 10% shift moves ω_c by $\sim 10\sigma$ against Planck); the neutrino sum $\Sigma m_\nu = R_2\, \omega_b\, (93.14)/(\tau_\Pi H_0)$, where KAL_0 appears *squared* through $R_2 = \Omega/(\mathrm{KAL}_0 T_r)$ and $\tau_\Pi H_0 = \mathrm{KAL}_0/(3 s_{\mathrm{DE}})$, so that $\mathrm{KAL}_0 \to 1.1\,\mathrm{KAL}_0$ gives $\Sigma m_\nu = 0.0566$ eV, *below* the oscillation floor 0.058 eV and therefore falsified; and Paper 10's screening functional. The cancellation above is thus local to $c^2_{s,\mathrm{eff}}$: τ_Π itself is anchored, through Σm_ν, to within roughly 10%.

Correction (2026-09-07). Earlier versions of this passage read "the same structural constant KAL_0 that normalises $K(X)$ also fixes the IS bulk viscosity". That is wrong about *which* $K(X)$: KAL_0 does not appear in the dark-energy action at all — Paper 7's kinetic function is $K = c_1 X + c_2 X^2$, and the X/KAL_0 form belongs to Paper 10's screening functional. Worse, the dark-energy action *cannot* supply a bulk viscosity: for shift-symmetric $K(X)$ the pressure perturbation is exactly adiabatic, $\delta p - c_s^2\, \delta\rho = 0$ identically in (c_1, c_2, X), so the entropy production of Eq. (6) vanishes and $\zeta = 0$. The IS layer is therefore a repair of the (w, c_s^2) *parametrisation* imposed by Boltzmann codes, not a property of the field; the sector's own prediction is the field value $c_s^2 = 0.021284$. The recurrence of KAL_0 in $\tilde{\zeta}$ is a resemblance across two different objects, not a derivation of one from the other, and is not claimed as one. See **OP-22b**.

1.3 The Israel–Stewart layer

The first-order (Eckart) relativistic fluid theory [Eckart, 1940] is acausal [Müller, 1967] and generically unstable [Hiscock and Lindblom, 1985]: perturbations propagate at infinite speed. Israel [Israel, 1976] and Israel & Stewart [Israel and Stewart, 1979] constructed a second-order causal theory in which dissipative fluxes satisfy relaxation equations with a finite relaxation time τ_Π. Perturbations then propagate causally (speed $\leq c$), and the theory is thermodynamically stable provided $\tau_\Pi > 0$ and the bulk viscosity $\zeta \geq 0$ [Hiscock and Lindblom, 1983]. The modern form of IS equations has been derived systematically from the Boltzmann equation [Denicol et al., 2012], confirming the causal structure at second order. Bulk-viscous cosmological models have been explored phenomenologically in Brevik and Grøn [2013], motivating the present IS dark-energy application.

SSEE identifies in Paper 1 a structural IS steady-state condition that fixes KAL_0 as the dimensionless viscosity, implying specific values of both ζ and τ_Π with no additional freedom. Earlier versions of this paper claimed that *these algebraically determined IS parameters exactly neutralise the gradient instability.* They neutralise it only for oscillations in the limit $\omega\tau_\Pi \to \infty$; the growing branch survives (Section 3). The claim is withdrawn, and the stability of the sector is assigned to the field.

1.4 Paper structure

Section 2 sets out the IS equations, the normalisation of the viscosity, and why the quasi-static reduction is the Navier–Stokes limit. Section 3 separates the high-frequency identity from stability and shows that the growing branch of the closure $c_s^2 = w$ survives the viscosity, while the field's sound speed removes it. Section 4 measures dark-energy clustering with the full IS system, including the passage through the phantom crossing. Section 5 gives the growth index, σ_8 and S_8. Section 6 places the IS layer in the EFT-of-dark-energy language, Section 7 discusses implications and collects the falsifiable predictions, and Section 8 concludes. Appendix A derives the dispersion relation and Appendix B gives the eigenvalues of the integrated system.

2 Israel-Stewart Theory for SSEE Dark Energy

2.1 Entropy production and the second-order IS equations

In relativistic thermodynamics the second law requires $\nabla_\mu S^\mu \geq 0$, where $S^\mu = su^\mu - q^\mu/T + \cdots$ is the entropy 4-current [Weinberg, 1971, Israel, 1976, Israel and Stewart, 1979]. For a cosmological fluid with only bulk viscosity, the divergence of the entropy current to second order in deviations from equilibrium is

$$\nabla_\mu S^\mu = \frac{1}{T}\left[-\Pi\,\theta - \frac{\tau_\Pi}{2\zeta T}\,\Pi\,\dot{\Pi} - \frac{\Pi^2}{\zeta}\right] \geq 0, \tag{6}$$

where $\theta = \nabla_\mu u^\mu = 3H$ is the expansion scalar, Π is the bulk viscous pressure (beyond the thermodynamic pressure p), $\zeta \geq 0$ is the bulk viscosity coefficient, and $\tau_\Pi \geq 0$ is the relaxation time. Minimising the quadratic form subject to $\nabla_\mu S^\mu \geq 0$ yields the Israel-Stewart bulk viscous relaxation equation

$$\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H - \frac{\tau_\Pi\,\Pi}{2}\left[\frac{\dot{\tau_\Pi}}{\tau_\Pi} - \frac{\dot{\zeta}}{\zeta} - \frac{\dot{T}}{T}\right], \tag{7}$$

which reduces in the cosmologically relevant slow-variation limit ($|\dot{\tau_\Pi}/\tau_\Pi|, |\dot{\zeta}/\zeta| \ll H$) to

$$\boxed{\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H.} \tag{8}$$

This is the form used throughout this paper. Two properties follow immediately from Eq. (6):

1. **Causality**: the IS relaxation gives a finite characteristic speed, $v_{\mathrm{sig}}^2 = c_{s,\mathrm{bare}}^2 + \zeta/[\tau_\Pi(\rho_{\mathrm{DE}} + p_{\mathrm{DE}})] \leq c^2$ [Hiscock and Lindblom, 1983] — the enthalpy again, as the inertia of the propagating mode.

2. **Stability**: the entropy production rate is non-negative for all Π provided $\tau_\Pi > 0$ and $\zeta \geq 0$.

Both conditions are satisfied in SSEE (Section 2.3).

2.2 SSEE IS parameters from algebraic structure constants

In the SSEE steady state $\dot{\Pi} = 0$, Eq. (8) gives $\Pi = -3\zeta H$. The dimensionless viscosity and relaxation time are defined by normalising to the dark-energy *enthalpy* and the Hubble rate:

$$\tilde{\zeta} \equiv \frac{\zeta\, H_0}{\rho_{\rm DE} + p_{\rm DE}} = \frac{\zeta\, H_0}{\rho_{\rm DE}(1+w_0)}, \tag{9}$$

$$\tau_\Pi H_0 \equiv \text{dimensionless IS relaxation time.} \tag{10}$$

The enthalpy $\rho + p$, not the density ρ, is the correct normalisation: it is the inertia that appears in the relativistic Euler equation (Appendix A, second line) and the origin of the $1/(1+w)$ factor multiplying the pressure gradient in the standard fluid perturbation equations. Normalising this way makes $\tilde{\zeta}/(\tau_\Pi H_0)$ directly a squared sound speed, with no residual factors.

Resolved: the normalisation of $\tilde{\zeta}$ (OP-22, closed 2026-09-06)

A previous version of this box reported $c^2_{s,\rm eff} = 0$ as an artefact of a normalisation mismatch between documents. That diagnosis was wrong, and we correct it here rather than leave it standing.

The question. $\tilde{\zeta} = \mathrm{KAL}_0/3$ is a dimensionless viscosity, but dimensionless *with respect to what*? The candidates differ by large factors, because one physical quantity $\zeta/(\rho\,\tau_\Pi)$ takes three values depending on the divisor:

divisor	$\zeta/(\rho\,\tau_\Pi)$
$\rho_{\rm crit}$ (total)	0.691119
$\rho_{\rm DE}$	1.000000
$\rho_{\rm DE} + p_{\rm DE}$ (enthalpy)	6.248039

The evidence, and it is not unanimous. Three places in the framework fix the normalisation, and one of them points the other way. The steady-state ansatz as originally written, $\Pi = -\mathrm{KAL}_0\,\rho_{\rm DE} H$, gives $\zeta = \mathrm{KAL}_0\rho_{\rm DE}/3$ and hence $c^2_{s,\rm eff} = w_0 + 5.248 = +4.41$: superluminal, and therefore not an option. The dispersion appendix of this paper requires the other convention.

The answer from inside this paper. The relativistic inertia of a sound wave is the enthalpy, so the dispersion relation of Appendix A carries $\zeta/[\tau_\Pi(\bar{\rho}+\bar{p})]$; it reduces to $\tilde{\zeta}/(\tau_\Pi H_0)$ *only if* $\tilde{\zeta}$ is enthalpy-normalised, $\zeta = \tilde{\zeta}\,(\rho_{\rm DE} + p_{\rm DE})/H_0$, which is what that appendix states. (Earlier versions cited a second witness, a friction coefficient ≈ 186 in the eigenvalue appendix; that number had no recorded computation and is withdrawn.)

The resolution, and why it is not merely a choice. We adopt the enthalpy form of the steady-state condition, $\Pi = -\mathrm{KAL}_0\,(\bar{\rho}+\bar{p})\,H$. The decisive argument is a limit test, independent of anything in SSEE. Take $w \to -1$ exactly. Then $\bar{\rho} + \bar{p} \to 0$: a cosmological constant carries no fluid degrees of freedom, so its bulk viscous

pressure *must* vanish. The enthalpy form gives $\Pi \to 0$ automatically; the $\rho_{\rm DE}$ form gives a finite viscous pressure for a cosmological constant, which is not a viable hydrodynamic limit. Two further reasons agree: it is what the dispersion appendix of this paper uses, and it is the form consistent with relativistic dissipative hydrodynamics, where the enthalpy is the inertia in the Euler equation and in the bulk-viscous transport alike [Maartens, 1996, Hiscock and Lindblom, 1985]. That the alternative is superluminal is then a consequence, not the reason for choosing. The archived note in which the $\rho_{\rm DE}$ form was first written is superseded by this.

Consequence. $c^2_{s,\rm eff} = w_0 + \tilde{\zeta}/(\tau_\Pi H_0) = w_0 + s_{\rm DE} = 0$ stands as the *high-frequency* sound speed of the IS fluid: subluminal, and not an artefact of normalisation. It is not a stability statement — the growing branch is treated in Section 3.

What is *withdrawn.* The *justification* of τ_Π given in the EFT appendix of Paper 1 [Almeida Vallejo, 2026a], which claimed to derive $\tau_\Pi H_0 = \mathrm{KAL}_0/(3s_{\rm DE})$ by saturating the causality bound written as $\zeta/(\rho_{\rm DE}\tau_\Pi) \leq 1$. That bound uses ρ where the enthalpy belongs; imposed correctly it returns $\tau_\Pi H_0 = \tilde{\zeta}(1+w_0) = 0.2946$, not 2.191. The factor between them is exactly $(1+w_0)^{-1} = 6.248$. Nothing numerical moves — $\tau_\Pi H_0 = \mathrm{KAL}_0\Omega/T_r$ is algebra, and was already stated in that form in Eq. (11) — but causality does not *derive* it, and Paper 1 no longer claims that it does.

Residual, still open — and smaller than it looked. The framework carries two sound speeds for the dark-energy sector:

mode	source	c_s^2
viscous fluid, $\omega\tau_\Pi \to \infty$ (this paper)	IS, $w_0 + s_{\rm DE}$	0
field, adiabatic	Paper 7 ghost condensate	0.021284

They are not in conflict in character — both describe a *clustered* dark energy, $c_s^2 \ll 1$. *Update (2026-09-07): they are not two channels.* A k-essence scalar carries exactly one propagating scalar degree of freedom (Paper 7, kinetic matrix $K_X + 2XK_{XX}$), and in the IS fluid the bulk viscous pressure Π is a relaxing auxiliary slaved to the density perturbation, not a second propagating mode. Both descriptions therefore count *one* mode, so the two numbers are two descriptions of the same wave and the field one is the fundamental of the pair: it comes from the action. The fluid description additionally borrows $c^2_{s,\rm ad} = w_0 < 0$, which is gradient-unstable on its own and is precisely what the viscosity "rescues" to 0 only at high frequency; the field needs no such rescue, returning +0.021284 directly. What remains open, and is now the whole of **OP-22b**, is narrower: the map from the field parameters to the effective fluid $(\tilde{\zeta}, \tau_\Pi)$ is not derived, so *why* the fluid limit lands exactly on 0 rather than on 0.021284 is not established (and, since the 0 is only a high-frequency limit, the physical value is the field's). The gap is the whole of Paper 7's prediction, so a measurement of c_s^2 remains sharp.

With the steady-state condition as currently written, and combined with the IS equation, one obtains [Almeida Vallejo, 2026a]:

$$\boxed{\tilde{\zeta} = \frac{\mathrm{KAL}_0\,\Omega}{M_v} = \frac{\mathrm{KAL}_0}{3} = 1.8405, \qquad \tau_\Pi H_0 = \frac{\mathrm{KAL}_0\,\Omega}{T_r} = 2.1912.} \tag{11}$$

Written this way the two are manifestly the *same* construction, built from the same

parents (KAL_0, Ω) and differing only in which sovereignty sits in the denominator: T_r (TRIAL) for the relaxation time, M_v (ATLAS) for the viscosity. The factor 3 in $\mathrm{KAL}_0/3$, which looks arbitrary in isolation, is simply $M_v = 3\Omega$. Both parameters are determined solely by $\mathrm{KAL}_0 = \beta + \pi$, $\Omega = \varphi + \pi$ and the sovereignties — no freedom remains. The ratio

$$\frac{\tilde{\zeta}}{\tau_\Pi H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3s_{\mathrm{DE}})} = s_{\mathrm{DE}} = |w_0| \tag{12}$$

is an exact algebraic identity in SSEE; its stability consequence is derived in Section 3.

2.3 Thermodynamic consistency

The IS theory is thermodynamically consistent if and only if:

(i) $\tau_\Pi > 0$: satisfied, $\tau_\Pi H_0 = 2.191 > 0$.

(ii) $\zeta \geq 0$: satisfied, $\tilde{\zeta} = 1.8405 > 0$, with the enthalpy of the field, $\rho_{\mathrm{DE}}(1 + w_0) > 0$, at all times.

(iii) The characteristic speed of the viscous mode, $v_{\mathrm{sig}}^2 = c_{s,\mathrm{bare}}^2 + \zeta/[\tau_\Pi(\rho_{\mathrm{DE}} + p_{\mathrm{DE}})] = w_0 + s_{\mathrm{DE}} = 0 \leq 1$: causal.

Earlier versions of this subsection divided by ρ_{DE} instead of the enthalpy, obtained $v_{\mathrm{sig}} = c$, and described the configuration as "maximally causal". With the enthalpy normalisation adopted above the viscous contribution alone is $s_{\mathrm{DE}} = 0.839950$ and the total characteristic speed is 0: subluminal, not saturating.

2.4 Linearised perturbation equations in Newtonian gauge

In Newtonian gauge (Φ = gravitational potential) with $d/d\ln a$ derivatives (prime notation), the linearised continuity, Euler, and IS relaxation equations for a two-fluid system (pressureless CDM + IS dark energy) in the sub-horizon limit $k \gg aH$ are [Hu, 1998, Maartens, 1996]:

$$\delta_m' = -\frac{k}{aH}\, v_m + 3\Phi', \tag{13}$$

$$v_m' = -v_m + \frac{k}{aH}\,\Phi, \tag{14}$$

$$\delta_{\mathrm{DE}}' = -(1 + w_0)\left[\frac{k}{aH}\, v_{\mathrm{DE}} - 3\Phi'\right], \tag{15}$$

$$v_{\mathrm{DE}}' = -(1 - 3c_s^2)\, v_{\mathrm{DE}} + \frac{k}{aH}\left[\frac{c_s^2\,\delta_{\mathrm{DE}}}{1 + w_0} + \tilde{\Pi} + \Phi\right], \tag{16}$$

$$\tau_\Pi H\, \tilde{\Pi}' = -\tilde{\Pi} - \tilde{\zeta}\,\frac{k}{aH}\,\frac{H}{H_0}\, v_{\mathrm{DE}}, \tag{17}$$

where $\tilde{\Pi} = \delta\Pi/[\rho_{\mathrm{DE}}(1 + w_0)]$ is the bulk viscous pressure perturbation normalised to the enthalpy of the field, and the Newtonian potential satisfies

$$k^2\,\Phi = -\frac{3}{2}\,\frac{H_0^2}{c^2}\left[\frac{\Omega_m\,\delta_m}{a} + a^2\,\Omega_{\mathrm{DE}}\, f_{\mathrm{DE}}(a)\,\delta_{\mathrm{DE}}\right], \tag{18}$$

with comoving k and $f_{\rm DE}(a) = \rho_{\rm DE}(a)/\rho_{\rm DE,0}$. The inertia of the dark-energy perturbation is the enthalpy of the *field*, $\rho_{\rm DE}(1+w_0) > 0$: in the closure of Paper 1 the field is assigned w_0 and the background viscous pressure the rest of $w(a)$, including the phantom crossing at $z_* = 0.314$ (an algebraic split: the field of Paper 7 rolls towards $w = -1$ and does not hold w_0 by itself, and that closure does not reproduce w_a dynamically — Paper 7, OP-24; we keep the present-day inertia and c_s^2, the latter only growing into the past) (Section 4.5 treats the total inertia, which changes sign there). Earlier versions used $1+w(a)$ clipped at 10^{-6} in the phantom era, wrote the gravitational terms of Eqs. (14)–(16) with the opposite sign — which made gravity repulsive in the integration — and divided the dark-energy one by $1+w$; all three are corrected. The $3\Phi'$ terms arise from the metric perturbation and are dropped in the sub-horizon integration, consistent with standard practice [Hu, 1998].

2.5 The quasi-static reduction is the Navier–Stokes limit

Earlier versions set $\tilde{\Pi}' = 0$ in Eq. (17), obtaining

$$\tilde{\Pi}_{\rm QS} = -\tilde{\zeta}\,\frac{k}{aH}\,\frac{H}{H_0}\,v_{\rm DE}, \tag{19}$$

which turns the viscosity into a velocity friction $\mathcal{F}(k,a) = (1-3c_s^2) + \tilde{\zeta}(k/aH)^2(H/H_0)$, and justified it by $\tau_\Pi H \gg 1$. The condition is the reverse. Dropping $\tau_\Pi H\,\tilde{\Pi}'$ requires the viscous pressure to relax faster than the perturbation evolves, $\tau_\Pi H \ll 1$; here $\tau_\Pi H \geq \tau_\Pi H_0 = 2.191$ at all times and grows into the past. Equation (19) is first-order (Navier–Stokes) viscosity, the acausal limit that Israel–Stewart was built to replace, and in that limit the viscosity only damps: it bounds the growth rate of the $c_s^2 = w_0$ branch at $1/(\tau_\Pi H_0) = 0.456$ per Hubble time, which is why the earlier integration showed nothing alarming. All results below integrate the full five-variable system (`LSODA`, relative tolerance 10^{-9}).

3 Q1: Gradient Stability — What the IS Identity Does and Does Not Do

3.1 IS dispersion relation from linear perturbation theory

For a relativistic viscous fluid with IS relaxation, a perturbation $\propto e^{i(kx-\omega t)}$ obeys (Appendix A)

$$\omega^2 = \left[c_{s,\rm bare}^2 + \frac{\zeta}{\tau_\Pi(\rho_{\rm DE}+p_{\rm DE})}\,\frac{-i\omega\tau_\Pi}{1-i\omega\tau_\Pi}\right]k^2. \tag{20}$$

In the limit $|\omega|\tau_\Pi \gg 1$ the viscous term becomes a pure stiffness and

$$\omega^2 = \left(c_{s,\rm bare}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\right)k^2 \equiv c_{s,\rm eff}^2\,k^2. \tag{21}$$

The full k-dependent interpolation between the Navier-Stokes ($k \to 0$) and IS ($k \to \infty$) limits is

$$c_{s,\rm eff}^2(k) = c_{s,\rm bare}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\cdot\frac{x^2}{1+x^2}, \quad x \equiv \frac{k\,c\,\tau_\Pi H_0}{aH}, \tag{22}$$

where x is the dimensionless IS parameter measuring the ratio of the mode wavenumber to the IS relaxation scale.

3.2 The high-frequency identity

High-frequency marginality (not stability)

In the SSEE dark energy framework, the IS parameters satisfy

$$\frac{\tilde{\zeta}}{\tau_\Pi H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3\,s_{\mathrm{DE}})} = s_{\mathrm{DE}} = |w_0|, \tag{23}$$

and therefore, for oscillations with $\omega\tau_\Pi \to \infty$*,*

$$c^2_{s,\mathrm{eff}} = c^2_{s,\mathrm{bare}} + |w_0| = w_0 + |w_0| = 0. \tag{24}$$

This is an exact algebraic identity within the SSEE constants, not a numerical coincidence. It expresses a construction-level consistency rather than a theorem constraining all IS models. It fixes the high-frequency sound speed of the fluid; it does not make the fluid stable (Section 3.3).

The identity follows directly from Eq. (12).

3.3 The growing branch: the closure $c_s^2 = w_0$ stays unstable

Stability is decided by modes that grow, not by modes that oscillate. For $e^{\lambda t}$ ($\omega = i\lambda$), Eq. (20) with $c^2_{s,\mathrm{bare}} = w_0 = -s_{\mathrm{DE}}$ gives

$$\lambda^2 = k^2\left[s_{\mathrm{DE}} - s_{\mathrm{DE}}\,\frac{\lambda\tau_\Pi}{1+\lambda\tau_\Pi}\right] = \frac{k^2\,s_{\mathrm{DE}}}{1+\lambda\tau_\Pi} > 0 \qquad \text{for every } k. \tag{25}$$

The viscous stiffness reaches s_{DE} only as $\lambda\tau_\Pi \to \infty$, so it never fully cancels the negative bare stiffness: at large k, $\lambda \simeq (k^2 s_{\mathrm{DE}}/\tau_\Pi)^{1/3}$, which is $3.2\,H_0$ at $k = 10\,H_0/c$ and $15.5\,H_0$ at $k = 100\,H_0/c$ — far faster than the expansion. The full cosmological integration agrees: with $c^2_{s,\mathrm{bare}} = w_0$ the ratio $\delta_{\mathrm{DE}}/\delta_m$ at $z = 0$ reaches 1.6×10^0 at $k = 20\,H_0/c$ and 2.4×10^2 at $k = 100\,H_0/c$ (`src/p05_IS/estabilidad_IS_completa.py`; its controls — $\zeta = 0$ returns $\lambda = k\sqrt{s_{\mathrm{DE}}}$, and the field's sound speed returns no growing mode — both pass).

What stabilises the sector is the field. Paper 7's ghost condensate has its own positive sound speed, $c_s^2 = +0.021284$. With it in place of w_0, Eq. (20) has no growing root, with or without viscosity, and the full integration gives $|\delta_{\mathrm{DE}}/\delta_m| = 1.5\times10^{-3}$ at $k = 10\,H_0/c$, falling as k^{-2} (Section 4). The closure $c_s^2 = w$ is the parametrisation a Boltzmann code imposes, not a property of the field (Section 1.2); the IS layer does not repair it, and it does not need to be repaired: the field never had the instability.

3.4 Uniqueness of the minimal-viscosity solution

The condition $c^2_{s,\mathrm{eff}} = 0$ requires $\tilde{\zeta}/(\tau_\Pi H_0) = |c^2_{s,\mathrm{bare}}|$. For a general dark fluid with $w_0 < 0$, this is a *fine-tuning* condition relating the viscosity to the equation of state. In SSEE both sides equal $s_{\mathrm{DE}} = |w_0|$, and the KAL_0 dependence cancels in the ratio $\tilde{\zeta}/(\tau_\Pi H_0) = (\mathrm{KAL}_0\Omega/M_v)/(\mathrm{KAL}_0\Omega/T_r) = T_r/M_v$. No number is tuned.

We are explicit about what does the work, because it is easy to overclaim here. The relaxation time $\tau_\Pi H_0 = \mathrm{KAL}_0\Omega/T_r$ is fixed independently in Paper 1 [Almeida Vallejo, 2026a], and $w_0 = -T_r/M_v$ is fixed independently in the background sector. Given those

two, requiring $c^2_{s,\mathrm{eff}} = 0$ leaves *no* freedom in $\tilde{\zeta}$: it must equal $|w_0|\,\tau_\Pi H_0 = \mathrm{KAL}_0\Omega/M_v$, with the T_r cancelling identically. So $\tilde{\zeta}$ is not an independent input.

What remains an input is the *requirement* that the high-frequency sound speed vanish, a selection principle rather than a fitted number. Earlier versions justified it as the smallest viscosity that removes the instability; Section 3.3 shows that no finite τ_Π removes it on the growing branch, so that justification is withdrawn and the principle is stated as what it is: the choice that makes the fluid's high-frequency limit match marginality.

3.5 Critical wavenumber and observational window

The transition between the Navier–Stokes and the IS regime of the oscillatory branch occurs at $x = 1$, defining

$$k_{\mathrm{crit}}(a) = \frac{aH}{c\,\tau_\Pi H_0}, \qquad \left.\frac{k_{\mathrm{crit}}}{H_0/c}\right|_{z=0} = \frac{1}{\tau_\Pi H_0} = 0.456. \tag{26}$$

Earlier versions read $k_{\mathrm{crit}} < H_0/c$ as "all sub-horizon modes are IS-stabilised". That reading concerned the oscillatory branch only; the growing branch of the closure $c_s^2 = w_0$ is unstable on both sides of k_{crit} (Section 3.3).

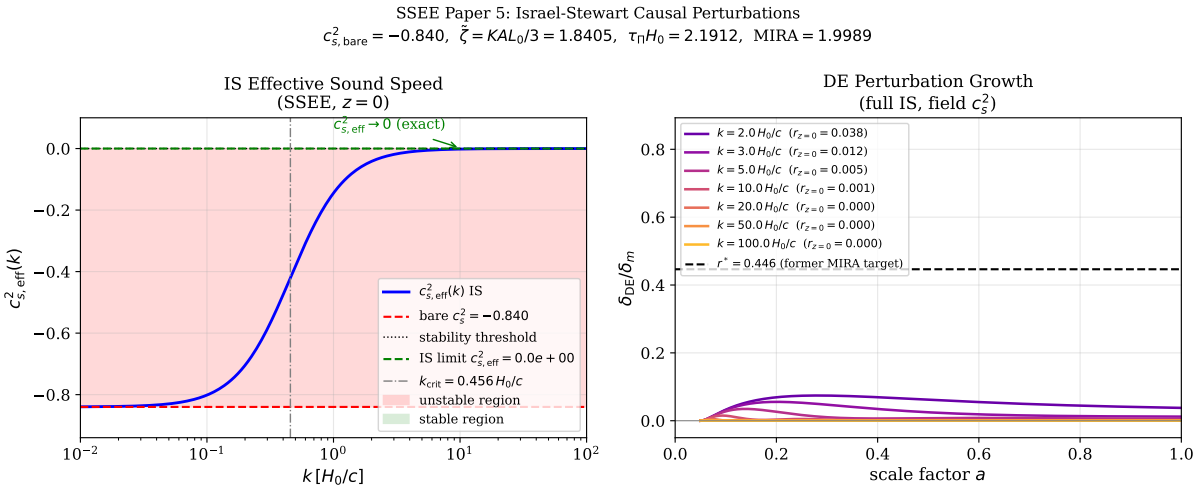


Figure 1: **Left:** IS effective sound speed $c^2_{s,\mathrm{eff}}(k)$ at $z = 0$ of the oscillatory branch (blue solid line), interpolating from the bare value $c_s^2 = w_0 = -0.839950$ (red dashed) at $k \ll k_{\mathrm{crit}}$ to the IS limit $c^2_{s,\mathrm{eff}} = 0$ (green dashed) at $k \gg k_{\mathrm{crit}}$. The IS transition is at $k_{\mathrm{crit}} = 0.456\,H_0/c$ (grey dash-dot). This curve is the high-frequency sound speed; it does not decide stability, which is set by the growing branch (Section 3.3). **Right:** Evolution of $\delta_{\mathrm{DE}}/\delta_m$ with scale factor a for representative wavenumbers $k/(H_0/c) = 2, 3, 5, 10, 20, 50, 100$, from the full IS integration with the field's sound speed $c_s^2 = 0.021284$; at $z = 0$ the ratio is 3.3×10^{-4} at $k = 20\,H_0/c$.

4 Q2: How Much Does Dark Energy Cluster?

4.1 Numerical setup and initial conditions

We integrate Eqs. (13)–(18) with $\tilde{\Pi}$ dynamical and the field's sound speed $c_s^2 = 0.021284$, from $a_{\mathrm{ini}} = 0.05$ ($z = 19$) to $a = 1$. At $z = 19$ the dark energy density fraction is

$\Omega_{\rm DE} f_{\rm DE}(a_{\rm ini})/(H/H_0)^2 = 1.9 \times 10^{-5}$ and the matter growing mode sets the initial conditions:

$$\delta_m(a_{\rm ini}) = a_{\rm ini}, \quad v_m(a_{\rm ini}) = -\frac{a_{\rm ini}\, H(a_{\rm ini})}{k}\, \delta_m(a_{\rm ini}), \quad \delta_{\rm DE} = v_{\rm DE} = \tilde{\Pi} = 0, \tag{27}$$

the velocity being the one for which $\delta_m' = \delta_m$ (earlier versions set $v_m = -a_{\rm ini} H/H_0$, too large by k/aH, which is why their $\delta_m(z=0)$ grew with k). The normalisation of δ_m cancels in all ratios (`src/p05_IS/ssee_paper5_IS_perturbations.py`).

4.2 The scale that a two-fold enhancement would need

The effective matter density seen through the Poisson equation is

$$\Omega_{m,\rm eff} = \Omega_m + \Omega_{\rm DE}\, r, \quad r \equiv \left.\frac{\delta_{\rm DE}}{\delta_m}\right|_{z=0}, \qquad \Omega_m = 0.308881, \quad \Omega_{\rm DE} = 0.691119, \tag{28}$$

with the *densities* in both slots (until 2026-09-26 it carried the saturations s_m and $s_{\rm DE}$), and we write $\mathcal{R} \equiv \Omega_{m,\rm eff}/\Omega_m$. Earlier versions posed $\mathcal{R}$ as a test of whether the perturbations reproduce MIRA = 1.9989, under a matter factor $\Omega_{m,\rm CMB} = \text{MIRA} \times s_m$ withdrawn on 2026-06-18 (OP-8). There is no factor of two to supply. The ratio a two-fold enhancement would have needed,

$$r^* = \frac{(\text{MIRA} - 1)\,\Omega_m}{\Omega_{\rm DE}} = \frac{0.998924 \times 0.308881}{0.691119} = 0.4464, \tag{29}$$

is kept only as the scale against which the measured r is small.

4.3 Numerical results

Table 1: Full linear IS integration from $z = 19$ to $z = 0$ with the field's sound speed. $\delta_m(z=0)$ for $\delta_m(z=19) = 0.05$; $r = \delta_{\rm DE}/\delta_m$; $\Omega_{m,\rm eff}$ via Eq. (28). The mean is over $k \geq 10\, H_0/c$.

$k\,[H_0/c]$	$\delta_m(z=0)$	$r = \delta_{\rm DE}/\delta_m$	$\Omega_{m,\rm eff}$	$\mathcal{R}$
2	0.7956	3.8×10^{-2}	0.3350	1.0847
3	0.7925	1.2×10^{-2}	0.3173	1.0271
5	0.7912	5.1×10^{-3}	0.3124	1.0115
10	0.7907	1.5×10^{-3}	0.3099	1.0033
20	0.7906	3.3×10^{-4}	0.3091	1.0007
50	0.7905	4.9×10^{-5}	0.3089	1.0001
100	0.7905	1.5×10^{-5}	0.3089	1.0000
Mean ($k \geq 10\, H_0/c$):				1.0010 ± 0.0013
Would require ($r = r^*$):				1.9989

The enhancement is $\mathcal{R} = 1.0010 \pm 0.0013$ at $k \geq 10\, H_0/c$: no enhancement. Its content is a bound on dark-energy clustering: $r = 1.5 \times 10^{-3}$ at $k = 10\, H_0/c$, falling to 1.5×10^{-5} at $k = 100\, H_0/c$, small and positive at every scale. The matter growth is the same at every k, as it must be for pressureless matter on sub-horizon scales. A measurement of

$\delta_{\rm DE}/\delta_m$ at the per-cent level at $k \gtrsim 10\,H_0/c$ and $z = 0$ would contradict the model. Earlier versions reported $\mathcal{R} = 0.9897 \pm 0.0167$ and a negative r bounded by 0.0175; those came from the integration with repulsive gravity, the growing-mode error in the initial velocity and the Navier–Stokes reduction, and are withdrawn.

4.4 Why the ratio falls as k^{-2}

With a positive sound speed the dark-energy perturbation below its sound horizon is a damped acoustic oscillation driven by the potential: the eigenvalues of the linearised system at $k = 10\,H_0/c$, $z = 0$ include the pair $-0.691 \pm 9.27\,i$ (Appendix B). Its forced amplitude is $\delta_{\rm DE} \sim (1 + w_0)\Phi/c_s^2$, and the Poisson equation (18) gives $\Phi \propto \delta_m/k^2$, so $r \propto k^{-2}$ — the scaling in Table 1.

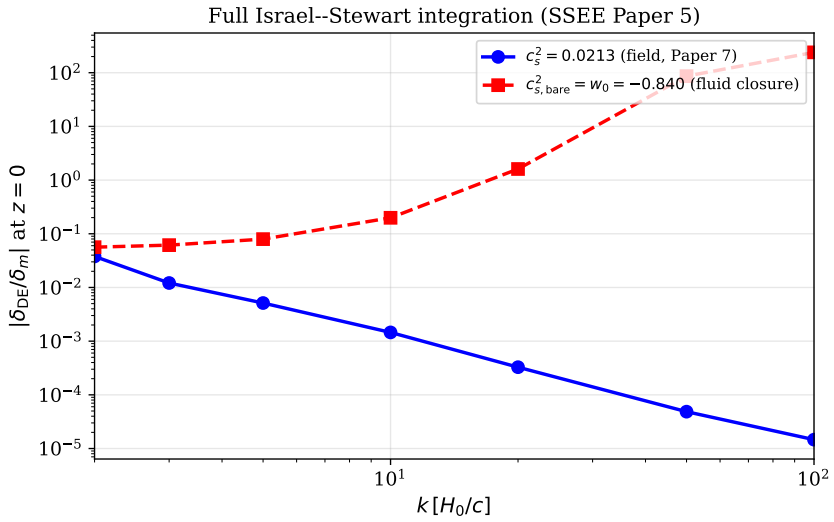


Figure 2: $|\delta_{\rm DE}/\delta_m|$ at $z = 0$ from the full Israel–Stewart integration. Blue: with the sound speed of the field of Paper 7, $c_s^2 = 0.021284$; the ratio falls as k^{-2} (Table 1). Red: with the fluid closure $c_{s,\rm bare}^2 = w_0$ and the same viscosity; the growing branch of Eq. (25) drives it to 2.4×10^2 at $k = 100\,H_0/c$. Earlier versions of this figure plotted $\mathcal{R}(k)$ against the withdrawn target MIRA.

4.5 Through the phantom crossing

The CPL history crosses $w = -1$ at $z_* = 0.314$ (Paper 1). The field cannot carry the crossing (its $c_s^2 < 0$ for $w < -1$, Paper 1); in the closure used here the background viscous pressure carries it, $1 + w = (1 + w_0)(1 - c)$ with $c = |\Pi|/[\rho_{\rm DE}(1 + w_0)]$. Two questions follow, and both are measured (`src/p05_IS/cruce_fantasma_IS.py`).

Do the perturbations cross? With the *total* inertia $1 + w(a)$, which vanishes at z_*, the rest-frame variables are singular. Written in $U = (1 + w)\theta/k$ with the pressure perturbation kept finite outside the rest frame [Kunz and Sapone, 2006], and with the field's sound speed, they remain finite: $\delta_{\rm DE}/\delta_m = -1.1 \times 10^{-1}$ at $k = 2\,H_0/c$ and -2.3×10^{-2} at $k = 10\,H_0/c$ at $z = 0$. The control with $c_s^2 = w_0$ reaches 8.2×10^3 at $k = 100\,H_0/c$.

Is the crossing within Israel–Stewart? The truncated (linear) theory is a near-equilibrium expansion, $|\Pi| \ll |p|$. Here $|\Pi|/|p| = |w_a|(1 - a)/|w_0|$ is 0.191 at the crossing, 0.399 at $z = 1$ and 0.558 at $z = 2.33$: beyond the linear theory for $z \gtrsim z_*$. That regime

is the one of the nonlinear extension of Maartens and Méndez [1997], in which phantom solutions driven by bulk viscosity have been constructed [Cruz and Lepe, 2017, Cruz et al., 2017]. The second law holds throughout, since $\zeta = \tilde{\zeta}\rho_{\rm DE}(1+w_0)/H_0 > 0$. Whether the nonlinear theory with the algebraic $(\tilde{\zeta}, \tau_\Pi)$ returns $w_a = -0.669975$ is the open part of the question (OP-24).

5 Q3: Growth Rate, σ_8, and the S_8 Tension

5.1 Linear growth factor and growth rate

The linear growth factor $D_1(a)$ satisfies, in the sub-horizon limit with dark-energy perturbations neglected ($|\delta_{\rm DE}/\delta_m| \leq 1.5 \times 10^{-3}$ for $k \geq 10\,H_0/c$, Section 4):

$$D_1'' + \left[2 + \frac{H'}{H}\right] D_1' = \frac{3}{2}\,\Omega_m(a)\,D_1, \tag{30}$$

where $\Omega_m(a) = \Omega_{m,\rm cosm} H_0^2/(H^2 a^3)$, $\Omega_{m,\rm cosm} = 0.308881$ is the gravitational background density (ω_m-direct $= \omega_m/h^2$, the CMB-inferred value; the earlier MIRA-enhanced value is superseded), and primes denote $d/d\ln a$. We define the growth rate

$$f(a) \equiv \frac{d\ln D_1}{d\ln a}, \tag{31}$$

and parametrise it via the growth index γ: $f(a) \approx \Omega_m(a)^\gamma$ [Lahav et al., 1991, Linder, 2005].

5.2 Numerical results for the SSEE growth index

We integrate Eq. (30) from $a = 10^{-3}$ to $a = 1$ (`DOP853`) with the SSEE Hubble rate. The growth index, from a least-squares fit of $\ln f = \gamma \ln \Omega_m(a)$ over $0 \leq z \leq 2$, is

$$\gamma_{\rm IS} = 0.5504 \pm 0.0003, \tag{32}$$

consistent with $\gamma_{\Lambda\rm CDM} = 0.55 \pm 0.01$ [Linder, 2005]. The growth index is indistinguishable from ΛCDM; with $\Omega_{m,\rm cosm} = 0.308881$ governing the Poisson source, the growth rate $f(z)$ itself is also very close to ΛCDM across all redshifts (Table 2).

Table 2: Growth rate $f = d\ln D_1/d\ln a$ at selected redshifts, comparing SSEE (IS dynamics, $\Omega_{m,\rm cosm} = 0.308881$) with ΛCDM ($\Omega_m = 0.315$). The growth index fit gives $\gamma_{\rm IS} = 0.5504$.

z	$\Omega_m^{\rm SSEE}(a)$	$f_{\rm SSEE}$	$f_{\Lambda\rm CDM}$	$\Delta f/f$
1.0	0.791	0.881	0.877	+0.4%
0.5	0.589	0.749	0.761	−1.6%
0.0	0.309	0.521	0.527	−1.3%

The SSEE and ΛCDM growth rates differ at the per-cent level (Table 2). With $\Omega_{m,\rm cosm} = 0.308881$ in the Poisson source, the gravitational clustering environment is close to ΛCDM ($\Omega_{m,\Lambda\rm CDM} = 0.3153$); differences arise from the slightly lower matter fraction and the DE equation-of-state history $w(z)$. Differences of this size are at the edge of what Stage IV surveys (DESI full shape, Euclid) will resolve in $f\sigma_8(z)$.

5.3 Linear growth factor and σ_8

The linear growth factor ratio relative to ΛCDM,

$$\mathcal{G} \equiv \frac{D_1^{\rm SSEE}(z=0)}{D_1^{\Lambda{\rm CDM}}(z=0)}, \tag{33}$$

is computed by normalising both growth factors to the same initial conditions at $a = 10^{-3}$ (matter-dominated epoch, $D_1 \propto a$). We find

$$\mathcal{G} = 1.0032. \tag{34}$$

SSEE grows *marginally faster* than ΛCDM (∼0.3%). Two background effects compete: $\Omega_{m,\rm cosm} = 0.308881$ is slightly *below* $\Omega_{m,\Lambda{\rm CDM}} = 0.3153$, which slows growth, while the CPL history ($w_0 = -0.839950$, $w_a = -0.669975$) has $w < -1$ for $z > z_*$, i.e. less dark energy than a cosmological constant in the past, which prolongs matter domination. The net is a small enhancement, below near-term weak-lensing growth sensitivity.

The σ_8 parameter is defined by $\sigma_8^2 = \int_0^\infty \frac{dk}{k}\,\Delta^2(k)\,W^2(kR_8)$, where $R_8 = 8\,h^{-1}\,$Mpc and $\Delta^2(k) \propto D_1^2(z=0)\,P(k)$ is the dimensionless matter power spectrum. Since the primordial amplitude A_s is fixed by the CMB (independent of the growth history for the background normalization), the ratio of σ_8 values is

$$\frac{\sigma_8^{\rm SSEE}}{\sigma_8^{\Lambda{\rm CDM}}} = \mathcal{G} = 1.0032, \tag{35}$$

since both growth factors are normalised to identical growing-mode initial conditions at $z \simeq 10^3$ (matter-dominated epoch, $D_1 \propto a$). Using $\sigma_{8,\Lambda{\rm CDM}} = 0.811$ [Planck Collaboration, 2020]:

$$\sigma_8^{\rm SSEE} = 0.811 \times 1.0032 = 0.8136 \pm 0.006, \tag{36}$$

the error being Planck's on σ_8, propagated. This IS linear-growth amplitude gives $S_8^{\rm IS} = 0.8136\sqrt{0.308881/0.3} = 0.8256$ (2.7σ from KiDS-1000), close to the conservative CLASS amplitude ceiling derived below.

5.4 S_8 prediction and the weak-lensing tension

The parameter $S_8 \equiv \sigma_8\,(\Omega_m/0.3)^{0.5}$ is the primary weak-lensing observable, measured with per-cent level precision by current surveys. Key measurements are:

$$S_8^{\rm Planck\,2018} = 0.832 \pm 0.013 \quad \text{[Planck Collaboration, 2020]}, \tag{37}$$

$$S_8^{\rm KiDS\text{-}1000} = 0.759 \pm 0.024 \quad \text{[Asgari et al., 2021]}, \tag{38}$$

$$S_8^{\rm DES\text{-}Y3} = 0.776 \pm 0.017 \quad (3{\times}2{\rm pt};\ \text{[Dark Energy Survey Collaboration, 2022]}). \tag{39}$$

Planck's ΛCDM ($\sigma_8 = 0.811$, $\Omega_m = 0.3153$, $S_8 = 0.8314$) lies 2.9σ from KiDS-1000 and 3.1σ from DES-Y3 with the same error treatment as below, one of the most persistent ΛCDM discrepancies [Weinberg et al., 2013].

For SSEE, the all-cold single sector uses the ω_m-direct CMB density $\Omega_{m,\rm CMB} = 0.308881$ ($= \omega_m/h^2$, no matter factor; the earlier MIRA value is superseded) and the CLASS amplitude ceiling $\sigma_8^{\rm SSEE} = 0.8149$:

$$\boxed{S_8^{\rm SSEE,\,ceiling} = 0.8149 \times \sqrt{\frac{0.308881}{0.3}} = 0.827.} \tag{40}$$

This single-sector ceiling is 0.36σ from Planck 2018, 2.74σ from KiDS-1000, and 2.82σ from DES-Y3 ($S_8 = 0.776 \pm 0.017$, 3×2pt). The all-cold limit therefore predicts *more* structure than the data require. A φ-DM two-sector was once proposed to resolve this; it is retracted in Paper 6. The ceiling quoted here is computed with A_s *fixed* to Planck, which imports the Planck–shear discrepancy; with A_s fixed instead to the model's own CMB fit, the same single sector predicts $S_8 = 0.8273$ against the raw KiDS-Legacy $\xi_\pm$ (0.8265 ± 0.0176 measured, Paper 6) and there is no challenge to resolve.

Table 3: S_8: the SSEE single-sector ceiling (A_s fixed to Planck, $\Omega_{m,\mathrm{CMB}} = 0.308881$, $\sigma_8 = 0.8149$) against observations, in units of the observational error (`p5_techo_sigma8_As_fijo.json`).

Dataset	S_8	$\lvert S_8^{\mathrm{SSEE}} - S_8\rvert/\sigma$
Planck 2018	0.832 ± 0.013	0.36
KiDS-1000	0.759 ± 0.024	2.74
DES-Y3	0.776 ± 0.017	2.82
SSEE (ceiling, this work)	0.827	—

With the same error treatment for both models, the IS linear-growth S_8 is $2.7\sigma/2.7\sigma$ from KiDS-1000/DES-Y3 and Planck's ΛCDM $2.9\sigma/3.1\sigma$: with A_s fixed to Planck the two are in the same position, SSEE marginally less tense because its Ω_m is lower. (Earlier versions quoted $3.0\sigma/3.3\sigma$ for ΛCDM using only the observational error for it and the combined error for SSEE; the asymmetry is corrected.) The falsification criterion of the growth sector is the one of Paper 6: a measurement placing S_8 more than 3σ from the prediction $S_8 = 0.8273$ made with A_s fixed to the model's own CMB fit.

Note on the amplitude convention, and on Paper 6. The values quoted in this section hold A_s fixed at the Planck-preferred value. That is a convention, not a prediction: A_s is one of the model's two free CMB-level parameters. Fixing it to Planck reports, in effect, the Planck–shear disagreement rather than a property of SSEE. Paper 6 instead confronts the *raw* KiDS-Legacy $\xi_\pm$ data with this same background held fixed and a *single* matter sector: the amplitude the shear asks for lies -0.25σ from the model's own CMB value, so A_s is fixed there too. The ϕ-DM two-sector split once described here is retracted. A baryonic-feedback figure of $S_8^{\mathrm{bar}} = 0.758$ was once quoted here; it is *retracted*. It applied the suppression factor of the matter power spectrum ($B_{\mathrm{eff}} = 0.9447$, a 5.53% suppression of $P(k)$) to the retired two-sector baseline, whereas the factor that acts on σ_8 itself is $B_{\sigma_8} = 0.9955$, a 0.45% suppression on this background. Applied to the single-sector ceiling, HMcode-2020 therefore leaves it at $S_8 \simeq 0.823$ — still a ceiling computed with A_s fixed to Planck, not a prediction. The single-sector value of this paper ($S_8 = 0.827$) is obtained with A_s fixed to Planck. With A_s fixed instead to the model's own CMB fit — no free cosmological parameter — SSEE predicts $S_8 = 0.8273$, 0.04σ from KiDS-Legacy (Paper 6); the two agree because the model's CMB amplitude lands on Planck's. (The earlier KiDS-1000 release, superseded, gave $S_8 = 0.7559 \pm 0.0189$ with A_s fitted.) Figure 3 shows these determinations on the same background.

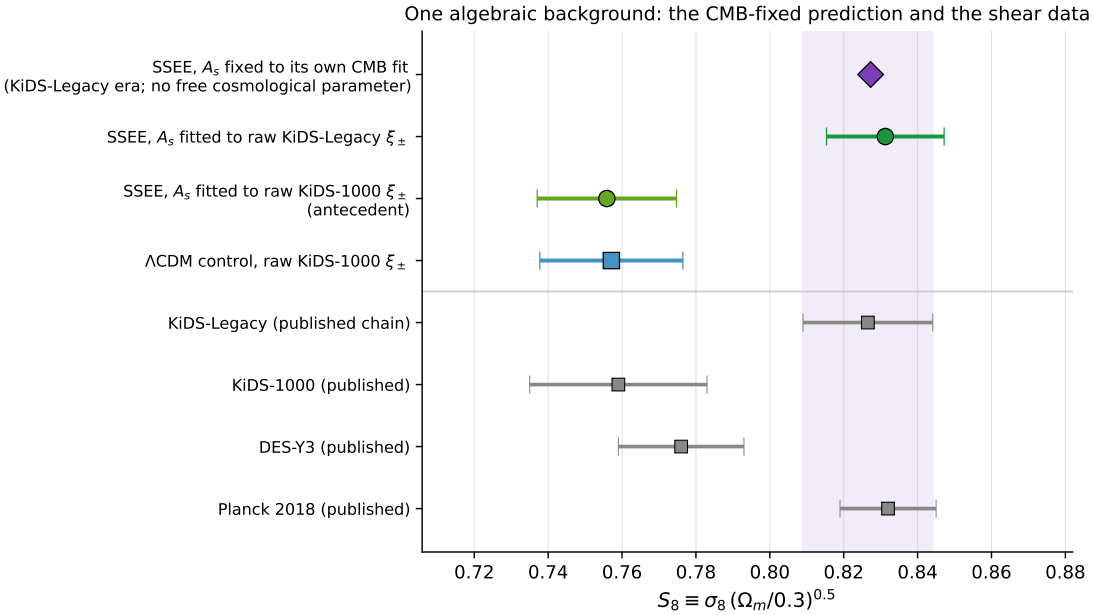


Figure 3: S_8 on the algebraic background of SSEE, identical in every SSEE row; only the dataset asked to fix A_s changes. Purple diamond: the prediction with A_s fixed to the model's own CMB fit — no free cosmological parameter — $S_8 = 0.8273$, 0.04σ from the KiDS-Legacy posterior (0.8265 ± 0.0176, purple band). Green: A_s fitted to the raw KiDS-Legacy $\xi_\pm$ (0.8313 ± 0.0159) and, as antecedent, to the raw KiDS-1000 $\xi_\pm$ (0.7559 ± 0.0189, 0.10σ from KiDS-1000). Blue: the ΛCDM control on the same raw KiDS-1000 data (0.7571 ± 0.0194). Grey: published KiDS-Legacy, KiDS-1000, DES-Y3 and Planck 2018 constraints with 1σ errors. The shift between the two KiDS releases is in the data, not in the model.

5.5 Comparison with RSD surveys: $f\sigma_8(z)$

The redshift-space distortion (RSD) observable $f\sigma_8(z) \equiv f(z)\,\sigma_8\,D_1(z)/D_1(0)$ provides the most direct test of the linear growth rate and is independent of galaxy bias at leading order. SSEE predicts $f\sigma_8(z)$ values essentially identical to ΛCDM at all redshifts, because with $\Omega_{m,\mathrm{cosm}} = 0.308881$ in the growth equation both models share the same gravitational clustering environment (Fig. 4).

Table 4 reports the predicted $f\sigma_8$ for both SSEE and ΛCDM at each survey redshift, together with the tension in units of σ_{obs}.

The mean tension of 0.70σ across six surveys is indistinguishable from the ΛCDM baseline (0.73σ). SSEE and ΛCDM produce virtually identical $f\sigma_8(z)$ curves when $\Omega_{m,\mathrm{cosm}} = 0.308881$ governs the Poisson source. The small difference in $\mathcal{G}$ (1.0032) is well below the precision of current surveys.

This result implies that $f\sigma_8$ measurements by Stage IV surveys (DESI full shape, Euclid) cannot distinguish SSEE from ΛCDM at the linear-growth level alone; distinguishing power must come from the background $H(z)$, already tested in Paper 2. (A transfer-function break at a ϕ-DM free-streaming scale was previously offered here as a second discriminant; it is withdrawn with the particle, Paper 6.) The DESI Year-1 full-shape power spectrum analysis [DESI Collaboration et al., 2025] measurements over $0.1 < z < 2.1$ are consistent with both ΛCDM and SSEE at this level of precision.

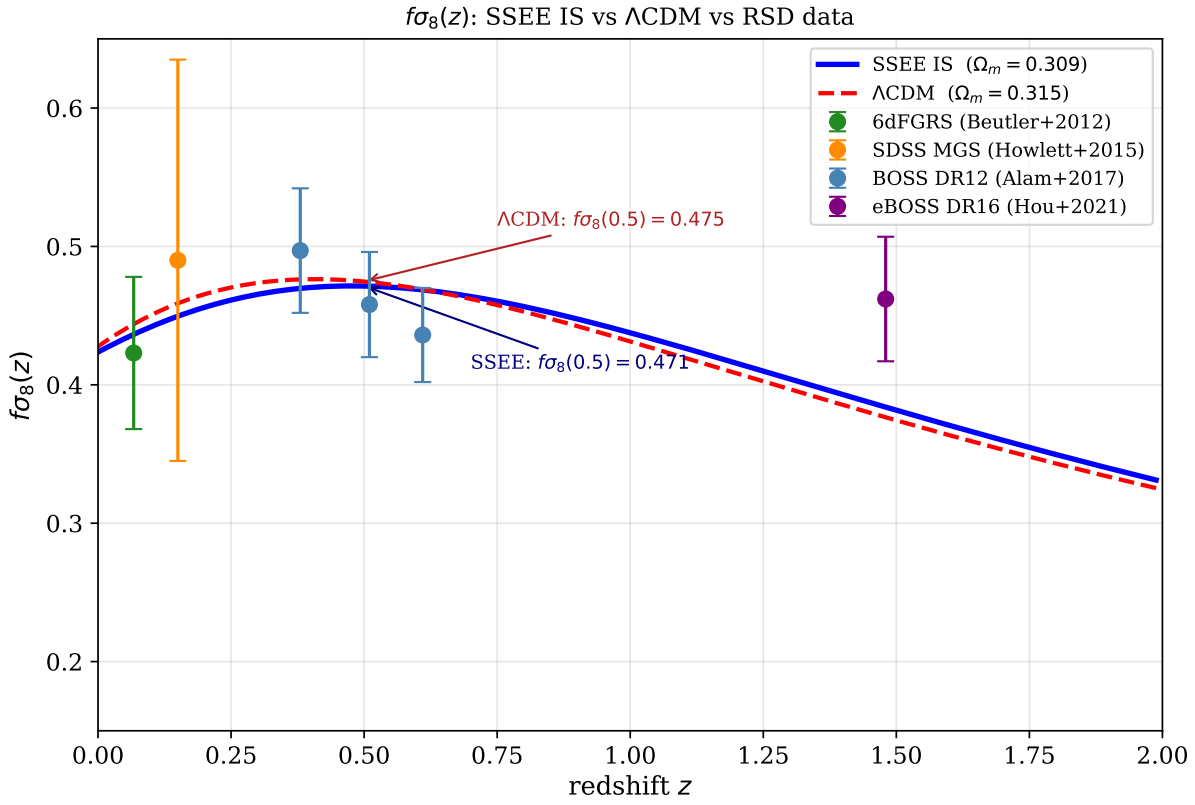


Figure 4: $f\sigma_8(z)$ for SSEE IS (solid blue) and ΛCDM (dashed red) versus RSD measurements from 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 [Alam et al., 2017], and eBOSS DR16 [Hou et al., 2021]. The SSEE prediction ($\Omega_{m,\mathrm{cosm}} = 0.308881$ in growth equation) is essentially identical to ΛCDM at $z < 1$, with a mean tension of 0.70σ across six data points (vs. 0.73σ for ΛCDM).

Table 4: $f\sigma_8$ tensions versus RSD surveys. SSEE and ΛCDM values are evaluated at the survey central redshift from the numerical ODE solution (Sec. 5.1). Tensions t use observational errors only ($t = |f\sigma_8^{\mathrm{pred}} - f\sigma_8^{\mathrm{obs}}|/\sigma_{\mathrm{obs}}$).

Survey	Ref.	z	$f\sigma_8^{\mathrm{obs}}$	$f\sigma_8^{\mathrm{SSEE}}$	t_{SSEE}	$t_{\Lambda\mathrm{CDM}}$
6dFGRS	Beutler et al. [2012]	0.067	0.423 ± 0.055	0.436	0.2σ	0.4σ
SDSS MGS	Howlett et al. [2015]	0.150	0.490 ± 0.145	0.449	0.3σ	0.2σ
BOSS DR12	Alam et al. [2017]	0.380	0.497 ± 0.045	0.470	0.6σ	0.5σ
BOSS DR12	Alam et al. [2017]	0.510	0.458 ± 0.038	0.471	0.3σ	0.4σ
BOSS DR12	Alam et al. [2017]	0.610	0.436 ± 0.034	0.468	1.0σ	1.0σ
eBOSS DR16	Hou et al. [2021]	1.480	0.462 ± 0.045	0.383	1.7σ	1.9σ
Mean tension					0.70σ	0.73σ

6 EFT Correspondence

In the effective field theory of dark energy [Horndeski, 1974, Gubitosi et al., 2013, Bellini and Sawicki, 2014] the action of Paper 7, $S = \int \sqrt{-g}\,[M_{\mathrm{Pl}}^2 R/2 + K(X)]$ with $K = c_1 X + c_2 X^2$, is a shift-symmetric k-essence: the only non-vanishing Bellini–Sawicki function is the kineticity α_K, with $\alpha_T = \alpha_B = \alpha_M = 0$, and the sound speed is $c_s^2 = 0.021284$

(Paper 7, verified there with `hi_class`). Two consequences for the IS layer:

- **No operator carries the viscosity.** For shift-symmetric $K(X)$ the pressure perturbation is exactly adiabatic, $\delta p - c_s^2 \delta\rho = 0$ (Section 1.2), so the action produces no bulk viscosity. The IS layer is a property of the fluid parametrisation used by Boltzmann codes and maps to no term of the action. Earlier versions identified $\tilde{\zeta}/(\tau_{\rm II} H_0)$ with $\bar{M}_2^2/M_{\rm Pl}^2$; that operator changes the tensor speed, $\alpha_T \neq 0$, which GW170817 excludes at $|\alpha_T| \lesssim 10^{-15}$ [LIGO Scientific Collaboration et al., 2017] and Paper 7 sets to zero. The mapping is withdrawn.

- **The ghost condensate is the SSEE sector, not an alternative.** Earlier versions contrasted SSEE with the ghost condensate [Arkani-Hamed et al., 2004] as a mechanism "with $c_s^2 = 1$". The ghost condensate has $c_s^2 = 0$ at its condensate point, and Paper 7's realisation, displaced from it to $X/X_{\rm min} = 1.045471$, has $c_s^2 = 0.021284$: it is the dark-energy sector of SSEE. The comparison table that accompanied the contrast is withdrawn with it.

7 Discussion

7.1 The withdrawn CMB-to-s_m ratio (historical)

Earlier versions asked whether the factor between $\Omega_{m,\rm CMB}$ and $1+w_0$ could arise from the IS viscous pressure at the background level, and reported a background integration giving an effective ratio of 3.04. No script or log of that integration survives, so the figure is withdrawn. The question itself no longer exists: the matter factor $\Omega_{m,\rm CMB} = \text{MIRA} \times s_m$ was withdrawn on 2026-06-18 when OP-8 closed, $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ is derived with no factor, and $s_m = 1 + w_0$ is a saturation, not a density.

The physical basis once attributed to Paper 6 is withdrawn, and the reason is worth recording because it is a result rather than a change of opinion. That construction defined a second matter sector by the difference $\Omega_{\phi\text{-DM}} = \Omega_{m,\rm CMB} - s_m = 0.308881 - 0.160050$, and drew from it an algebraically fixed particle mass and a free-streaming cutoff. But $s_m = 0.160050$ is not a density: it is $1 + w_0$, a quantity of the equation of state. The subtraction is dimensionally well formed and physically empty — it takes a measured density minus a number belonging to the equation of state — so the sector it defined had nothing to be made of. The near-equality of $\Omega_{m,\rm CMB}/s_m$ with MIRA, once read as an echo, is a ratio between objects of different kinds and carries no information.

Independently, the tension that motivated the second sector does not survive contact with raw data. It was measured against the compressed statistic S_8, which is derived under ΛCDM assumptions, and with A_s fixed to Planck. Against the raw KiDS-Legacy $\xi_\pm$, with A_s fixed to the model's own CMB fit and a single matter sector, the model predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured, with no free cosmological parameter (Paper 6).

The consequence for this paper is confined to §7.1: MIRA retains its algebraic value $(3\varphi + \pi)/4 = 1.998924$ and its role in the screening fraction $f_{\rm screen}$ of Paper 9, but it has *no proposed physical carrier at present*. We state that as an open gap rather than fill it. The Q1, Q2 and Q3 results of this paper are unaffected: none of them uses the second sector.

7.2 Implications for JWST early galaxy counts

Dark-energy perturbations effectively vanish at sub-horizon scales in this model, because the field's positive sound speed turns them into damped acoustic oscillations below the sound horizon, with $\delta_{\rm DE}/\delta_m \propto k^{-2}$ (1.5×10^{-3} at $k = 10\,H_0/c$; §4.4). On that smooth background the spherical-collapse threshold is $\delta_c^{\rm SSEE} = 1.67634$ at $z_c = 0$ and 1.68647 at $z_c = 10$, against ΛCDM's 1.67599 → 1.68646: the two agree to the third decimal. The value $\delta_c^{\rm SSEE} = 1.6284$ carried by earlier versions was a postulate of Paper 4, now **withdrawn** (OP-27); the model's own dynamics do not produce it. Matter clusters in a background with $\Omega_m = 0.308881$ — the one matter density of the model; the number 0.160050 that earlier versions of this subsection placed here is $1 + w_0$, an equation-of-state parameter, and never was a density (Paper 1, § *One matter density*).

Recomputed, with the amplitude convention made explicit. Earlier versions of this subsection reported $n_{\rm SSEE}/n_{\Lambda\rm CDM} \approx 1.4$–$2.0$ for $M > 10^{11}\,M_\odot$ at $z > 10$ from a Press–Schechter calculation whose growth factor was fed $\Omega_m = 0.160050$ as a matter density. That input was a category error — 0.160050 is $1 + w_0$ — and the calculation has been redone (`src/p02_mcmc/ssee_press_schechter.py`, 2026-09-07) with $\Omega_m = 0.308881$, the measured growth index $\gamma_{\rm IS} = 0.5504$ (§5.2) and the global rate $H_0^{\rm glob} = 67.962$. With both cosmologies evaluated at their *own* CMB-fitted amplitude ($\sigma_8 = 0.81530$ for SSEE, 0.811 for ΛCDM) and with the *derived* collapse threshold:

z	$M = 10^{11} M_\odot$	$M = 10^{12} M_\odot$
10	0.990	0.945
12	0.984	0.921
15	0.972	0.878

There is no enhancement, and the sign is the opposite of what earlier versions claimed. With the threshold the model actually produces, SSEE forms marginally *fewer* massive early haloes than ΛCDM. The deficit grows with mass and redshift, reaching 0.892 at $3 \times 10^{12} M_\odot$, $z = 10$ and 0.774 at the same mass at $z = 15$, and it is not an amplitude effect: SSEE carries the *larger* σ_8. It is the growth history — a smaller Ω_m (0.308881 against 0.3153) and the CPL background grow less between $z \sim 10$ and today. At the masses JWST actually probes ($M \sim 10^{10.8} M_\odot$) the ratio is 0.99: indistinguishable from ΛCDM. SSEE offers no account of the JWST excess, and the 1.4–2.0 of earlier versions was an artefact of the withdrawn δ_c.

The amplitude convention is not a detail. This ratio compares two cosmologies and is only meaningful if both are evaluated at the same epoch's amplitude. Giving SSEE the amplitude the earlier, superseded fit to raw KiDS-1000 shear preferred ($\sigma_8 = 0.7449$) while leaving ΛCDM at Planck's 0.811 inverts the result to a deficit (0.823 at $10^{11} M_\odot$ and 0.346 at $10^{12} M_\odot$, $z = 10$). That inversion is not a statement about collapse physics: it is the CMB-versus-late-time amplitude disagreement entering through the back door, and it would affect ΛCDM identically if the same mismatch were imposed on it. High-redshift abundance comparisons must therefore state which amplitude each cosmology is given; matched-amplitude is what is reported above.

7.3 The S_8–$f\sigma_8$ alignment and the role of $\Omega_{m,\mathrm{cosm}}$

With the correct cosmological background $\Omega_{m,\mathrm{cosm}} = 0.308881$ governing the Poisson source in the growth equation, both S_8 and $f\sigma_8$ predictions align closely with ΛCDM:

- S_8 **(weak-lensing amplitude):** The IS linear-growth estimate is $S_8 = 0.8256 \pm 0.006$, $2.7\sigma/2.7\sigma$ from KiDS-1000/DES-Y3, against $2.9\sigma/3.1\sigma$ for Planck's ΛCDM with the same error treatment; the conservative CLASS amplitude ceiling is $S_8 = 0.827$ (2.74σ KiDS-1000). Both numbers hold A_s *fixed* to Planck, which imports the Planck–shear discrepancy; with A_s fixed instead to the model's own CMB fit, the raw KiDS-Legacy $\xi_\pm$ give $S_8 = 0.8273$ predicted against 0.8265 ± 0.0176 measured (Paper 6). There is no weak-lensing tension for the single-sector IS framework to resolve.

- $f\sigma_8$ **(RSD growth rate):** SSEE predicts $f\sigma_8(z = 0.5) = 0.471$, matching the BOSS DR12 value of 0.458 ± 0.038 at $+0.35\sigma$. The mean tension across six RSD surveys is 0.70σ, essentially identical to ΛCDM (0.73σ).

The two observables are consistent: once $\Omega_{m,\mathrm{cosm}} = 0.308881$ sets the gravitational background, SSEE differs from ΛCDM only through the slightly lower matter fraction and the DE equation-of-state history $w(z)$ with $w_0 = -0.839950$, $w_a = -0.669975$. This drives a ∼0.3% enhancement of $\mathcal{G}$ and σ_8, which translates to a marginally different S_8.

The apparent weak-lensing tension does not survive contact with the raw data. A two-sector ϕ-DM construction was once proposed to remove it by free-streaming suppression; Paper 6 retracts that construction and shows the tension was an artefact of holding A_s at its Planck value. With A_s fixed to the model's own CMB fit, the single sector predicts $S_8 = 0.8273$ against the 357 raw KiDS-Legacy $\xi_\pm$ points (0.8265 ± 0.0176 measured).

7.4 The H_0 tension and SSEE

In SSEE the global rate is the output of the SH0ES cascade, $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) = 67.962 \pm 0.968$ km s^{-1} Mpc^{-1} (the measured local rate is the input; the pure number $3(\varphi + \pi)^2$ is the comparison target, not an input [Almeida Vallejo, 2026e]), consistent with Planck 2018 (67.36 ± 0.54) at 0.54σ (errors combined) and with DESI DR2. The IS layer does not enter the screening cascade of Paper 9, so nothing in this paper bears on the H_0 comparison; and, as Section 5 shows, it does not move σ_8 or S_8 either.

7.5 Causal structure and Hiscock-Lindblom stability

The IS theory satisfies the Hiscock–Lindblom criteria [Hiscock and Lindblom, 1983] — non-negative entropy production and causal propagation — provided $\tau_\Pi > 0$ and $\zeta \geq 0$, both of which hold here (§2.3); the characteristic speed is $v_{\mathrm{sig}}^2 = w_0 + s_{\mathrm{DE}} = 0 \leq c^2$. Those criteria concern the equilibrium state and the characteristic cone; they do not by themselves forbid a growing mode, and Section 3.3 shows that the closure $c_s^2 = w_0$ has one. Earlier versions stated that the signal speed "saturates the causality bound" and added that Eckart theories would produce instabilities in the CMB at $\ell > 1000$; the first came from the ρ-normalisation abandoned above and the second had no calculation behind it. Both are withdrawn.

Table 5: Falsifiable predictions of SSEE IS dark energy, with observational targets and current/future datasets capable of testing each prediction. A single failure at the stated level would falsify SSEE.

Prediction	SSEE value	Test level	Dataset / Timeline
c_s^2 of the dark-energy field (Paper 7; the IS high-frequency limit is 0, OP-22b)	0.021284	not yet measured	CMB lensing cross-correlations [Calabrese et al., 2011] / forecast
S_8, A_s fixed to the model's own CMB fit (Paper 6)	0.8273 (no free cosmological parameter)	$> 3\sigma$ (now 0.04σ from KiDS-Legacy)	KiDS-Legacy / current; Euclid / 2027
$\gamma_{\rm IS}$	0.5504 ± 0.0003	γ degenerate with ΛCDM at current precision	DESI full shape, Euclid / 2026–2028
$f\sigma_8(z = 0.5) = 0.471$	0.471	$+0.35\sigma$ BOSS; mean 0.70σ (6 surveys)	DESI full-shape / 2026
D_1 ratio $\mathcal{G}$	1.0032	below near-term sensitivity	Euclid weak lensing / 2027
DE clustering $\delta_{\rm DE}/\delta_m$ at $z = 0$	1.5×10^{-3} at $k = 10\,H_0/c$, $\propto k^{-2}$	per-cent level excluded	Future 21cm surveys / 2030+
MIRA $= (3\varphi + \pi)/4$ (algebraic; the two-sector basis is retracted, Paper 6 — see OP-8)	1.998924	enters only the local screening fraction $f_{\rm screen}$ (Paper 9), not Ω_m	local H_0 ladders (SH0ES, CCHP) / current

7.6 Falsifiable predictions summary

8 Conclusions

We have performed a linear Israel-Stewart causal viscous perturbation analysis of SSEE dark energy in the sub-horizon regime, addressing three physically motivated questions.

Q1 — Gradient stability. The SSEE structure constants satisfy $\tilde{\zeta}/(\tau_\Pi H_0) = s_{\rm DE} = |w_0|$, so the high-frequency sound speed of the IS fluid vanishes, $w_0 + s_{\rm DE} = 0$. That is not stability: on the growing branch the viscous stiffness never reaches $s_{\rm DE}$, and the closure $c_s^2 = w_0$ grows at every k ($\lambda = 15.5\,H_0$ at $k = 100\,H_0/c$). The dark-energy sector is stable

because the field of Paper 7 has $c_s^2 = +0.021284$; with it no mode grows. The quasi-static reduction of earlier versions was the Navier–Stokes limit and hid the growing branch.

Q2 — Dark-energy clustering. The full IS integration with the field's sound speed gives $\mathcal{R} = 1.0010 \pm 0.0013$ at $k \geq 10\, H_0/c$ and $\delta_{\rm DE}/\delta_m = 1.5 \times 10^{-3}$ at $k = 10\, H_0/c$, falling as k^{-2}: a bound on dark-energy clustering. The perturbations cross the phantom divide at $z_* = 0.314$ without divergence; the viscous pressure that carries the crossing exceeds the near-equilibrium range of linear Israel–Stewart for $z \gtrsim z_*$ ($|\Pi|/|p| = 0.399$ at $z = 1$), and whether the nonlinear theory returns w_a is open (OP-24). MIRA has no carrier in this sector and survives only as the algebraic invariant entering $f_{\rm screen}$ (Paper 9).

Q3 — Growth rate and S_8 (SSEE tracks ΛCDM closely). The IS growth index $\gamma_{\rm IS} = 0.5504 \pm 0.0003$ is numerically indistinguishable from $\gamma_{\Lambda\rm CDM} \simeq 0.55$. With $\Omega_{m,\rm cosm} = 0.308881$ governing the Poisson source, the growth factor ratio is $\mathcal{G} = 1.0032$, giving $\sigma_8^{\rm SSEE} = 0.8136 \pm 0.006$ and an IS linear-growth $S_8^{\rm IS} = 0.8256 \pm 0.006$ (2.7σ from KiDS-1000, against 2.9σ for Planck's ΛCDM with the same error treatment). The conservative CLASS amplitude ceiling, $S_8^{\rm ceiling} = 0.827$ (2.74σ KiDS-1000), coincides with the IS linear-growth estimate above: once massive neutrinos are included, the two independent routes agree. The mean $f\sigma_8$ tension across six RSD surveys is 0.70σ, essentially identical to ΛCDM (0.73σ). There is no weak-lensing tension left for the IS sector to resolve: against the raw KiDS-Legacy $\xi_\pm$, with A_s fixed to the value the model's own CMB fit supplies, a single matter sector with no free cosmological parameter predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured; the earlier, superseded fit to KiDS-1000 with A_s free gave $S_8 = 0.7559 \pm 0.0189$, 0.10σ (Paper 6). The $S_8 = 0.8256$ quoted above is a ceiling obtained with A_s fixed to Planck, which imports the Planck–KiDS offset rather than predicting it. The ϕ-DM construction that once appeared here as the required resolution is retracted with the particle.

Together, these results establish that SSEE dark energy is: (i) stable, through the field's positive sound speed rather than through the IS layer, which is causal and thermodynamically consistent but does not by itself remove the instability of the closure $c_s^2 = w$ (Q1), (ii) predictive of a growth history essentially identical to ΛCDM once $\Omega_{m,\rm cosm} = 0.308881$ governs the growth equation (Q3), with $f\sigma_8$ tension of 0.70σ (vs. 0.73σ for ΛCDM), and (iii) weakly clustered, $\delta_{\rm DE}/\delta_m \propto k^{-2}$ (Q2), with no MIRA factor to derive; the physical basis previously attributed to Paper 6's two-sector ϕ-dark matter construction is **withdrawn**: that construction and its particle are retracted (Paper 6). MIRA's algebraic value $(3\varphi + \pi)/4$ is unchanged and it survives in the model only through the screening fraction $f_{\rm screen}$ (Paper 9); **what remains open is what, if anything, provides its physical basis** — an explicitly declared gap, not a filled one. The primary falsifiable prediction in the growth sector is $S_8 = 0.8273$, obtained with A_s fixed to the value the model's own CMB fit supplies and no free cosmological parameter, against 0.8265 ± 0.0176 from the raw KiDS-Legacy $\xi_\pm$ (Paper 6); a future release placing S_8 more than 3σ from it falsifies the growth sector. The two-sector WDM correction once quoted here is retracted.

Data Availability

The Python scripts `src/p05_IS/ssee_paper5_IS_perturbations.py` (Q1–Q3, growth rate and S_8), `estabilidad_IS_completa.py` (growing branch) and `cruce_fantasma_IS.py` (phantom crossing), their logs, and all output figures are available in the SSEE repository at https://github.com/mikealmeida1721/SSEE-Cosmologia (Zenodo DOI: 10.5281/zenodo.19654118). All computations use publicly available Python packages (NumPy, SciPy, Matplotlib) and, for the amplitude ceiling, CLASS.

Acknowledgements

This research was carried out as an independent investigation. The author thanks the open-source communities behind NumPy, SciPy, Matplotlib, and CAMB, and acknowledges the Planck, DESI, KiDS, and DES teams for making their data publicly available.

A Derivation of the IS Dispersion Relation

We derive Eq. (21) from the linearised IS equations in Minkowski space (the cosmological curvature terms are subdominant in the sub-horizon limit).

Linearised IS system. Write $\rho = \bar{\rho} + \delta\rho$, $p = \bar{p} + \delta p$, $\Pi = \bar{\Pi} + \delta\Pi$, $u^\mu = (1, \mathbf{v})$ with $|\mathbf{v}| \ll 1$. The continuity and Euler equations give:

$$\dot{\delta\rho} + (\bar{\rho} + \bar{p})\,\nabla\cdot\mathbf{v} = 0, \tag{41}$$

$$(\bar{\rho} + \bar{p})\,\dot{\mathbf{v}} + \nabla(\delta p + \delta\Pi) = 0. \tag{42}$$

The IS relaxation equation linearises to:

$$\tau_\Pi\,\dot{\delta\Pi} + \delta\Pi = -\zeta\,\nabla\cdot\mathbf{v}. \tag{43}$$

Plane-wave ansatz. For modes $e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$:

$$\delta\Pi = -\frac{i\zeta k}{1 - i\omega\tau_\Pi}\,v_\parallel, \tag{44}$$

where $v_\parallel = \mathbf{k}\cdot\mathbf{v}/k$ is the longitudinal velocity.

Effective sound speed. Combining with the continuity and Euler equations — whose inertia is the enthalpy $\bar{\rho} + \bar{p}$, as written above, *not* $\bar{\rho}$:

$$\omega^2 = \left[c_{s,\mathrm{bare}}^2 + \frac{\zeta}{\tau_\Pi(\bar{\rho}+\bar{p})}\,\frac{-i\omega\tau_\Pi}{1 - i\omega\tau_\Pi}\right] k^2. \tag{45}$$

For $|\omega|\tau_\Pi \gg 1$ the bracket tends to $c_{s,\mathrm{bare}}^2 + \zeta/[\tau_\Pi(\bar{\rho}+\bar{p})]$. With the SSEE normalisation $\zeta = \tilde{\zeta}\,(\rho_{\mathrm{DE}} + p_{\mathrm{DE}})/H_0$ (Section 2.2) this is

$$c_{s,\mathrm{eff}}^2 = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}, \tag{46}$$

recovering Eq. (21). For a growing mode, $\omega = i\lambda$, the factor $-i\omega\tau_\Pi/(1 - i\omega\tau_\Pi)$ becomes $\lambda\tau_\Pi/(1 + \lambda\tau_\Pi) < 1$, which gives Eq. (25). (Earlier versions wrote $\zeta = \tilde{\zeta}(\rho + p)H_0$ and equated $\tilde{\zeta}H_0/\tau_\Pi$ with $\tilde{\zeta}/(\tau_\Pi H_0)$; the placement of H_0 is corrected, and the result is unchanged.)

B Eigenvalue Analysis of the IS Perturbation Matrix

The five-variable system $\mathbf{Y} = [\delta_m, v_m, \delta_{\rm DE}, v_{\rm DE}, \tilde{\Pi}]^\top$ of Eqs. (13)–(18), with the field's sound speed, is linear; its Jacobian with respect to $\ln a$, evaluated by the same code that integrates it at $z = 0$ and $k = 10\,H_0/c$ (`p5_IS_perturbations_Q2.json`, `autovalores_k10_z0`), has eigenvalues

$$\lambda = +0.346, \quad -0.011, \quad -0.691 \pm 9.27\,i, \quad -1.345. \tag{47}$$

The positive one is the matter growing mode at frozen coefficients and the most negative the decaying one; the complex pair is the damped acoustic oscillation of the dark-energy perturbation, which carries the k^{-2} suppression of Section 4.4. No dark-energy mode grows. Earlier versions printed four eigenvalues for a different matrix with no recorded computation; they are withdrawn.

References

M. S. Abdul-Karim et al. DESI DR2 results II: Measurements of baryon acoustic oscillations and cosmological constraints. *Phys. Rev. D*, 112:083515, 2025.

S. Alam et al. The clustering of galaxies in the completed SDSS-III baryon oscillation spectroscopic survey: cosmological analysis of the DR12 galaxy sample. *Monthly Notices of the Royal Astronomical Society*, 470:2617–2652, 2017. doi: 10.1093/mnras/stx721.

Mike Edison Almeida Vallejo. A minimal-parameter framework for cosmological dynamics and galaxy cluster mass discrepancies via structural self-energy expansion (SSEE). *arXiv*, 2026a. SSEE Paper 1.

Mike Edison Almeida Vallejo. UV extension of SSEE dark energy: $m^4 = 45\alpha^2\rho_{\rm crit}$ and a conditional self-consistency check on the Hubble tension. *arXiv*, 2026b. SSEE Paper 10.

Mike Edison Almeida Vallejo. Bayesian MCMC validation of the structural self-energy expansion framework (SSEE): Model comparison against ΛCDM and CPL using DESI DR2, Planck 2018, and galaxy-cluster mass data. *arXiv*, 2026c. SSEE Paper 2.

Mike Edison Almeida Vallejo. SSEE and the CMB power spectrum: Acoustic peak reproduction via the ω_m-direct CMB matter density. *arXiv*, 2026d. SSEE Paper 3.

Mike Edison Almeida Vallejo. Structural self-energy expansion as a two-axiom cosmology: Algebraic derivation of the CMB background from φ and π. *arXiv*, 2026e. SSEE Paper 4.

N. Arkani-Hamed, H.-C. Cheng, M. A. Luty, and S. Mukohyama. Ghost condensation and a consistent infrared modification of gravity. *Journal of High Energy Physics*, 2004: 074, 2004. doi: 10.1088/1126-6708/2004/05/074.

C. Armendariz-Picon, V. Mukhanov, and P. J. Steinhardt. Essentials of k-essence. *Physical Review D*, 63:103510, 2001. doi: 10.1103/PhysRevD.63.103510.

M. Asgari et al. KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics. *Astronomy & Astrophysics*, 645:A104, 2021. doi: 10.1051/0004-6361/202039070.

Emilio Bellini and Ignacy Sawicki. Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity. *JCAP*, 07:050, 2014. doi: 10.1088/1475-7516/2014/07/050.

F. Beutler et al. The 6df galaxy survey: baryon acoustic oscillations and the local hubble constant. *Monthly Notices of the Royal Astronomical Society*, 423:3430–3444, 2012. doi: 10.1111/j.1365-2966.2012.21136.x.

I. Brevik and Ø. Grøn. Universe models with negative bulk viscosity. *Astrophysics and Space Science*, 347(2):399–404, 2013. doi: 10.1007/s10509-013-1525-z.

E. Calabrese, D. N. Spergel, B. D. Sherwin, and R. Nagata. Non-adiabatic dark energy perturbations. *Physical Review D*, 83:023011, 2011. doi: 10.1103/PhysRevD.83.023011.

R. R. Caldwell. A phantom menace? cosmological consequences of a dark energy component with super-negative equation of state. *Physics Letters B*, 545:23–29, 2002. doi: 10.1016/S0370-2693(02)02589-3.

M. Chevallier and D. Polarski. Accelerating universes with scaling dark matter. *International Journal of Modern Physics D*, 10:213–224, 2001. doi: 10.1142/S0218271801000822.

M. Cruz, N. Cruz, and S. Lepe. Phantom solution in a non-linear Israel–Stewart theory. *Phys. Lett. B*, 769:159, 2017.

N. Cruz and S. Lepe. Crossing the phantom divide with dissipative normal matter in the Israel–Stewart formalism. *Phys. Lett. B*, 2017. doi: 10.1016/j.physletb.2017.01.035.

Dark Energy Survey Collaboration. Dark Energy Survey year 3 results: Cosmological constraints from galaxy clustering and weak lensing. *Physical Review D*, 105:023520, 2022. doi: 10.1103/PhysRevD.105.023520.

G. S. Denicol, H. Niemi, E. Molnar, and D. H. Rischke. Derivation of transient relativistic fluid dynamics from the Boltzmann equation. *Physical Review D*, 85:114047, 2012. doi: 10.1103/PhysRevD.85.114047.

DESI Collaboration, A. G. Adame, et al. DESI 2024 V: Full-Shape Galaxy Clustering from Galaxies and Quasars. *J. Cosmol. Astropart. Phys.*, 2025(09):008, 2025. doi: 10.1088/1475-7516/2025/09/008.

C. Eckart. The thermodynamics of irreversible processes. III. relativistic theory of the simple fluid. *Physical Review*, 58:919–924, 1940. doi: 10.1103/PhysRev.58.919.

G. Gubitosi, F. Piazza, and F. Vernizzi. The effective field theory of dark energy. *Journal of Cosmology and Astroparticle Physics*, 2013(02):032, 2013. doi: 10.1088/1475-7516/2013/02/032.

W. A. Hiscock and L. Lindblom. Stability and causality in dissipative relativistic fluids. *Annals of Physics*, 151:466–496, 1983. doi: 10.1016/0003-4916(83)90288-9.

W. A. Hiscock and L. Lindblom. Generic instabilities in first-order dissipative relativistic fluid theories. *Physical Review D*, 31:725–733, 1985. doi: 10.1103/PhysRevD.31.725.

G. W. Horndeski. Second-order scalar-tensor field equations in a four-dimensional space. *International Journal of Theoretical Physics*, 10:363–384, 1974. doi: 10.1007/BF01807638.

J. Hou et al. eBOSS collaboration: The completed SDSS-IV extended baryon oscillation spectroscopic survey: BAO and RSD measurements from anisotropic clustering analysis of the quasar sample between redshift 0.8 and 2.2. *Monthly Notices of the Royal Astronomical Society*, 500:1201–1221, 2021. doi: 10.1093/mnras/staa3234.

C. Howlett, A. J. Ross, L. Samushia, W. J. Percival, and M. Manera. The clustering of the SDSS main galaxy sample – II. mock galaxy catalogues and a measurement of the growth of structure from redshift space distortions at $z = 0.15$. *Monthly Notices of the Royal Astronomical Society*, 449:848–866, 2015. doi: 10.1093/mnras/stu2693.

W. Hu. Structure formation with generalized dark matter. *The Astrophysical Journal*, 506:485–494, 1998. doi: 10.1086/306274.

W. Israel. Nonstationary irreversible thermodynamics: A causal relativistic theory. *Annals of Physics*, 100:310–338, 1976. doi: 10.1016/0003-4916(76)90064-6.

W. Israel and J. M. Stewart. Transient relativistic thermodynamics and kinetic theory. *Annals of Physics*, 118:341–372, 1979. doi: 10.1016/0003-4916(79)90130-1.

M. Kunz and D. Sapone. Crossing the phantom divide. *Phys. Rev. D*, 74:123503, 2006.

O. Lahav, P. B. Lilje, J. R. Primack, and M. J. Rees. Dynamical effects of the cosmological constant. *Monthly Notices of the Royal Astronomical Society*, 251:128–136, 1991. doi: 10.1093/mnras/251.1.128.

LIGO Scientific Collaboration, Virgo Collaboration, Fermi GBM, and INTEGRAL. Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A. *Astrophys. J. Lett.*, 848:L13, 2017.

E. V. Linder. Exploring the expansion history of the universe. *Physical Review Letters*, 90:091301, 2003. doi: 10.1103/PhysRevLett.90.091301.

E. V. Linder. Cosmic growth history and expansion history. *Physical Review D*, 72: 043529, 2005. doi: 10.1103/PhysRevD.72.043529.

R. Maartens. Causal thermodynamics in relativity. *arXiv*, 1996.

R. Maartens and V. Méndez. Nonlinear bulk viscosity and inflation. *Phys. Rev. D*, 55: 1937, 1997.

I. Müller. Zum paradoxon der wärmeleitungstheorie. *Zeitschrift für Physik*, 198:329–344, 1967. doi: 10.1007/BF01326412.

Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.

D. H. Weinberg, M. J. Mortonson, D. J. Eisenstein, C. Hirata, A. G. Riess, and E. Rozo. Observational probes of cosmic acceleration. *Physics Reports*, 530:87–255, 2013. doi: 10.1016/j.physrep.2013.05.001.

S. Weinberg. Entropy generation and the survival of protogalaxies in an expanding universe. *The Astrophysical Journal*, 168:175–194, 1971. doi: 10.1086/151073.